

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 3: Class Test**

**COURSE NAME: COMPUTER VISION**

**COURSE CODE: ITA05**

**COURSE OUTCOME COVERED**

**CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)**

**Assessment Tool Weightage: 10%**

**Code of Conduct:**

***I PAVAN JANA(192472313) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.***

***I understand that any violation of academic integrity rules will result in disciplinary action.***

**Signature of the student: pavanj**

**Rubrics for Evaluation**

| <i>Criteria</i>                  | <i>Weightage</i> | <i>Level 4 – Exemplary</i>                                         | <i>Level 3 – Proficient</i>                    | <i>Level 2 – Developing</i>                         | <i>Level 1 – Beginning</i>                      | <i>Marks(100)</i> |
|----------------------------------|------------------|--------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------|-------------------------------------------------|-------------------|
| <i>Understanding of Concepts</i> | <i>25%</i>       | <i>Demonstrates thorough, accurate understanding of concepts</i>   | <i>Good understanding with minor gaps</i>      | <i>Basic understanding with some misconceptions</i> | <i>Poor understanding; major misconceptions</i> |                   |
| <i>Explanation</i>               | <i>25%</i>       | <i>Detailed, well-explained answers with relevant examples</i>     | <i>Clear explanation with adequate details</i> | <i>Limited explanation; lacks depth</i>             | <i>Poor or unclear explanation</i>              |                   |
| <i>Application</i>               | <i>20%</i>       | <i>Effectively applies concepts with strong analytical ability</i> | <i>Applies concepts with minor errors</i>      | <i>Limited application with weak analysis</i>       | <i>No application or incorrect analysis</i>     |                   |
| <i>Organization</i>              | <i>15%</i>       | <i>Well-structured answers with logical flow and coherence</i>     | <i>Organized with minor issues in flow</i>     | <i>Basic structure, lacks clarity</i>               | <i>Poor organization, incoherent answers</i>    |                   |

|                 |            |                                                                  |                                                    |                                                     |                                                       |  |
|-----------------|------------|------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|--|
| <b>Accuracy</b> | <b>10%</b> | <b><i>Completely accurate and covers all required points</i></b> | <b><i>Mostly accurate with minor omissions</i></b> | <b><i>Partially correct with missing points</i></b> | <b><i>Incorrect answers with major gaps</i></b>       |  |
| <b>Clarity</b>  | <b>5%</b>  | <b><i>Clear, neat, professional presentation</i></b>             | <b><i>Understandable with minor issues</i></b>     | <b><i>Limited clarity, readability issues</i></b>   | <b><i>Poorly written, difficult to understand</i></b> |  |

**1. An image contains high-frequency noise.Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.**

**Answer:**

The appropriate filter is a Low-Pass Filter (LPF), such as an Ideal Low-Pass Filter (ILPF), Butterworth Low-Pass Filter (BLPF), or Gaussian Low-Pass Filter (GLPF).

**Working:**

- High-frequency noise is represented by high-frequency components in the frequency domain.
- A Low-Pass Filter allows low-frequency components to pass while attenuating or removing high-frequency components.
- The image is transformed to the frequency domain using the Fourier Transform, the LPF is applied, and the filtered image is transformed back using the Inverse Fourier Transform.

**Justification:**

- Since noise mainly exists in the high-frequency region, suppressing these frequencies reduces the noise effectively.

**Impact on Image Quality:**

- Noise is significantly reduced.
- The image becomes smoother.
- Fine details and sharp edges may be slightly blurred depending on the cutoff frequency.

**2. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.**

**Answer:**

**A Gaussian Low-Pass Filter (GLPF)** is suitable for this task.

**Working:**

- The Gaussian filter provides a smooth transition between passed and blocked frequencies.
- It preserves low-frequency components that contain the main image information while gradually suppressing high-frequency noise.

**Application:**

1. Convert the image to the frequency domain.
2. Apply the Gaussian Low-Pass Filter.
3. Perform the inverse Fourier transform to reconstruct the filtered image.

**How it achieves the result:**

- Low-frequency components representing overall brightness and object structures are retained.
- High-frequency noise is gradually reduced without introducing ringing artifacts.

**Result:**

- Reduced noise.
- Better preservation of image quality compared to the Ideal Low-Pass Filter.

**3. An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.**

**Answer:**

**A High-Pass Filter (HPF)** is used for edge enhancement.

**Working:**

- Edges correspond to high-frequency components.
- The High-Pass Filter suppresses low-frequency information and retains high-frequency components.
- After inverse transformation, edges become more prominent.

**Justification:**

- Since edge information lies in high-frequency regions, preserving these frequencies enhances object boundaries.

**Effect on Frequency Components:**

- Low frequencies are attenuated.
- High frequencies are emphasized.

**Result:**

- Sharper edges.
- Improved visibility of fine details.
- Noise may also become more noticeable if present.

**4.A smooth version of an image is required.Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.**

**Answer:**

A **Low-Pass Filtering technique** is used for smoothing.

**Working:**

- Transform the image into the frequency domain.
- Apply a Low-Pass Filter to remove high-frequency components.
- Perform the inverse Fourier transform.

**Justification:**

- High-frequency components represent noise and abrupt intensity changes.
- Removing them produces a smoother image.

### Results:

- Noise is reduced.
- Image appears smoother.
- Some loss of edge sharpness may occur.

**5. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.**

### Answer:

The suitable technique is Threshold Segmentation (Global Thresholding).

### Working:

- Select a threshold value  $T$ .
- Pixels with intensity greater than  $T$  are classified as the object.
- Pixels below  $T$  are classified as the background.

$$g(x,y) = \begin{cases} 1, & f(x,y) \geq T \\ 0, & f(x,y) < T \end{cases}$$

### Justification:

- When the object is much brighter than the background, thresholding provides accurate separation.

### Effectiveness:

- Simple and computationally efficient.
- Produces accurate segmentation when intensity contrast is high.

**6.You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.**

**Answer:**

The suitable method is Otsu's Thresholding.

**Working:**

- Otsu's method automatically calculates the optimal threshold from the image histogram.
- It minimizes intra-class variance while maximizing inter-class variance.

**Application:**

1. Compute the histogram.
2. Determine the optimal threshold.
3. Separate foreground and background.

**Result:**

- Automatic segmentation.
- No manual threshold selection required.
- Effective for images with bimodal histograms.

**7.An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.**

**Answer:**

The **Canny Edge Detection** algorithm is suitable.

**Working:**

1. Smooth the image using a Gaussian filter.
2. Compute gradient magnitude and direction.
3. Apply Non-Maximum Suppression.
4. Perform Double Thresholding.

5. Use Edge Tracking by Hysteresis.

**Justification:**

- Produces thin and accurate edges.
- Removes noise before edge detection.

**Effectiveness:**

- High detection accuracy.
- Good localization.
- Low false edge detection.

**8.Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.**

**Answer:**

The appropriate technique is **Region Growing Segmentation**.

**Working:**

- Select one or more seed pixels.
- Compare neighboring pixels with the seed.
- If neighboring pixels have similar intensity, they are added to the region.
- Continue until no more similar pixels remain.

**Application:**

- Choose seed points inside the object.
- Apply similarity criteria.
- Expand the region.

**Reasoning:**

- Pixels belonging to the same object generally have similar intensity values.

**Result:**

- Connected regions representing objects are obtained.

**9. Detected edges in an image appear broken.**

**Explain and justify a method to improve edge continuity, including its role in post-processing.**

**Answer:**

The suitable method is **Morphological Closing**.

**Working:**

- Closing consists of **Dilation followed by Erosion**.
- Dilation fills small gaps between edge segments.
- Erosion restores the original edge thickness while maintaining connectivity.

**Justification:**

- Broken edges reduce object recognition accuracy.
- Closing connects nearby edge fragments.

**Role in Post-processing:**

- Improves edge continuity.
- Removes small gaps.
- Produces smoother object boundaries.

**10. You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively.**

**Answer:**

The appropriate technique is **Boundary Extraction using Morphological Operations**.

**Working:**

1. Perform erosion on the binary image.
2. Subtract the eroded image from the original image.

$\text{Boundary} = \text{Original} - \text{Eroded}$

**Application:**

- Convert the image into binary form.
- Apply erosion.
- Subtract the eroded image from the original.

**How it identifies boundaries:**

- Erosion removes outer pixels of the object.
- Subtracting the eroded image leaves only the object's outline.

**Result:**

- Accurate extraction of object boundaries.
- Preserves the shape of the object.
- Useful for object recognition and shape analysis.